- **Ideation work**

  The core idea of my AR project, AR Forecast, originated from the desire to make weather data more interactive and interesting. I believed I could change the everyday experience of information by removing forecasts from flat screens and bringing them into the real world through augmented reality. Rather than seeing only a weather app, users could view weather conditions in front of them in any environment, even indoors with an AR interface on a device.

  Weather forecasting is worldwide and real-time, providing up-to-the-second information to facilitate AR use. I pictured the user floating above the sky or on a plane surface, seeing the 3D scene of mixed cloud cover, amount of rain, thunderstorms, and fog, based on the forecast. This will be entertaining and educational.

- **Pre-production work**

The pre-production process of AR Forecast included establishing visual composition, becoming familiar with the sources of information, and creating a plan as to how weather data in real-time would be translated into an engaging AR experience.

First, I sketched out the simple AR screen layout. Sketched what the 3D weather objects like clouds, rain and sun would appear as when placed into the real world with AR. These sketches enabled me to decide where objects used in unity would be placed. I also pictured how different types of weather would be represented, like 3D clouds for cloudy conditions, or raindrop particles for rainy. These drawings acted as a rough storyboard and helped guide my asset creation and Unity scene setup.

To drive real-time weather changes in the AR world, I considered using APIs. I chose to use the Open-Meteo API because it offered free, easily accessible weather data, such as current weather data. During this stage, I used debug method in c# and in Unity confirm the message in console to check if I was able to fetch the data correctly. I chose the fields that I would use, for instance:

- ✓ Time
- ✓ Temperature (Celsius)
- ✓ Relative Humidity (%)
- ✓ Is Day or Night
- ✓ Weather code (wmo code)
- ✓ Rain (mm)
- ✓ Cloud cover for visual cloudiness (%)

External data within this research was essential in organizing how the external data would be associated with visuals in Unity AR.

To achieve a realistic representation, I studied how the weather is normally graded in the UK. I applied two core aspects:

Rainfall: I studied that rain is normally graded by amount in terms of hourly rates:

- Light rain: under 0.5 mm/hour
- Moderate rain: between 0.5 to 4 mm/hour
- Heavy rain: over 4 mm/hour

These were operationally and functionally defined using information from the Met Office's Hazard Manager tool and National Meteorological Library and Archive .

Cloud cover: I found that cloudiness is normally defined as:

- Low: 0–30%
- Medium: 30–70%
- High: 70–100%

This classification was possible because of Meteoblue's cloud variable guidelines that showed me how percentage coverages of the clouds relate to their actual conditions of the sky. This research guided me in my application of percentages in scaling clouds within my AR scene from thin scattered to overcast.

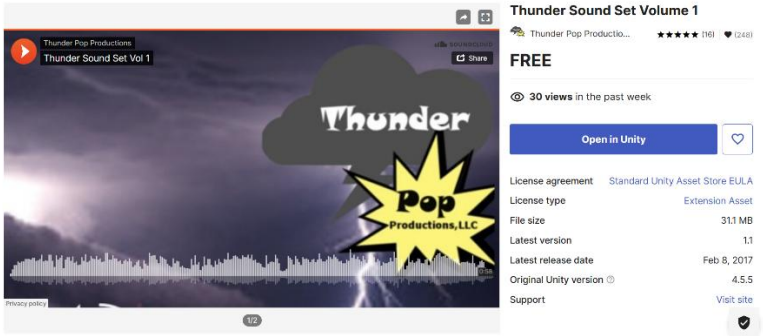
- **References**

Met Office (no date). *Using the map and accessing the weather layers (beta)* [PDF]. Available from: https://www.metoffice.gov.uk/binaries/content/assets/metofficegovuk/pdf/services/government/using-the-map-and-accessing-the-weather-layers-beta.pdf [Accessed 30 April 2025].

Met Office (no date). *factsheet_3-water-in-the-atmosphere* [PDF]. Available from: https://www.metoffice.gov.uk/binaries/content/assets/metofficegovuk/pdf/research/library-and-archive/library/publications/factsheets/factsheet_3-water-in-the-atmosphere.pdf [Accessed 30 April 2025].

meteoblue (no date) Clouds [online]. Available from: https://content.meteoblue.com/en/research-education/specifications/weather-variables/clouds[Accessed 30 April 2025].

- **Inventory of Unity assets used**



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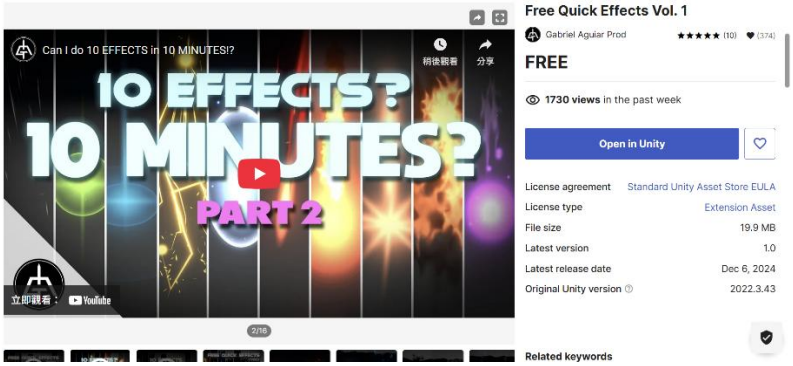

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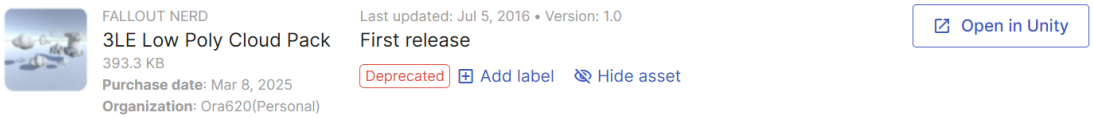
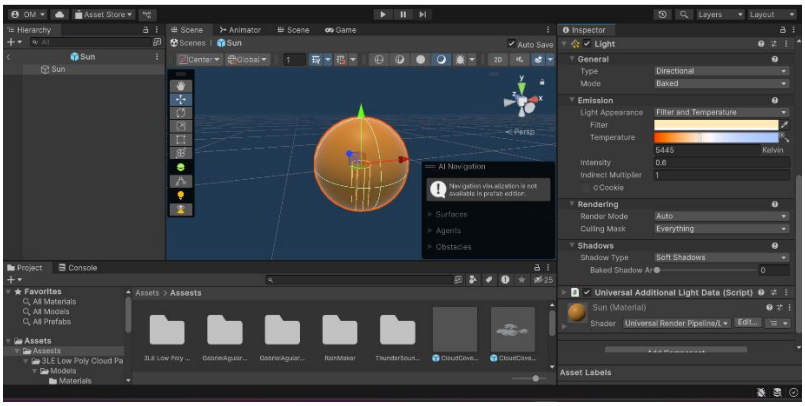
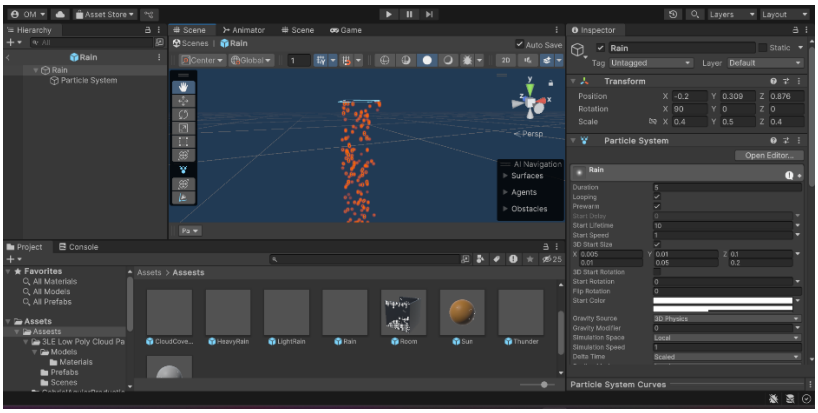


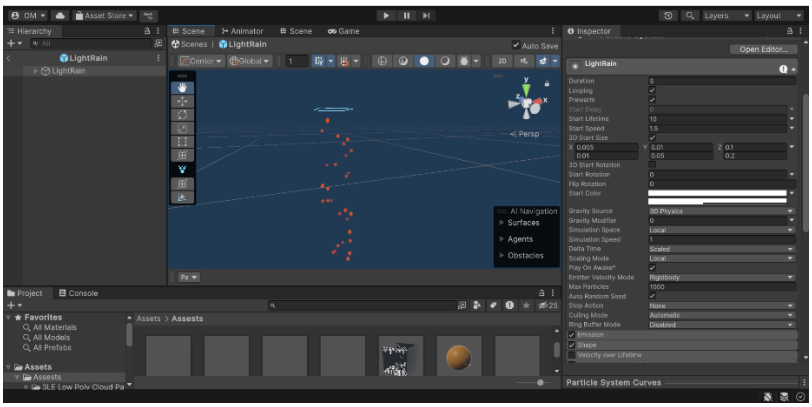
https://assetstore.unity.com/packages/vfx/particles/environment/rain-maker-2d-and-3d-rain-particle-system-for-unity-34938

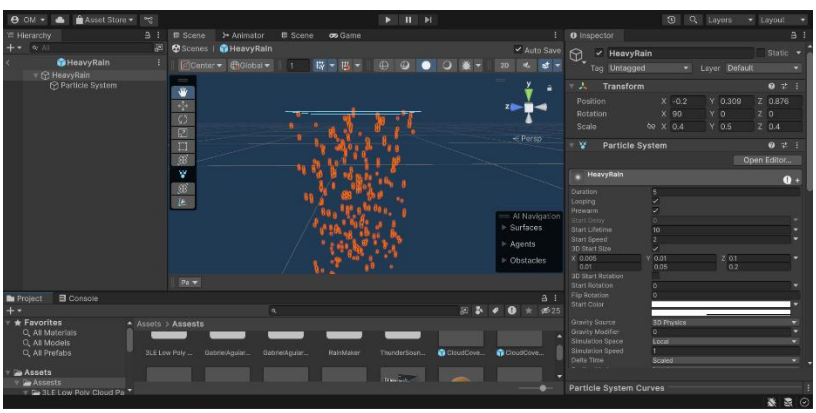
- **Inventory of any assets that created**



By following tutorial video found from asset store:







- **500 words Evaluation / reflection**

Creating the AR Forecast project was an extremely difficult yet learning experience. It allowed me to apply novel problem-solving and technical execution to create an augmented reality experience showing real-time weather data. Rather than concentrating on user interaction in the form of image tracking, I concentrated on mashing up external data into an AR setting. This course led me to explore how data might be employed constructively to enhance visual information to aid in user understanding of their surroundings in the most positive manner.

The project was of great significance to using real-time weather data onboard utilizing the Open-Meteo API. I mastered calling APIs, dealing with JSON data in Unity, and taking data points to visual markers within AR. Yes, this took some learning to get started on, but honestly, I'd say C# wasn't near as terrible as I'd previously thought. By utilizing a simple script and variable naming convention, I prevented confusion and reduced errors. Employing the “if…else if” code paradigm to construct conditional logic came second nature after comprehending that it's essentially programming like English. These discoveries convinced me more in my coding proficiency, and I began to appreciate the problem-solving nature of programming.

One of the biggest things I learned was the importance of debugging and organization of clean code. At the beginning of the project, I had minor issues that were hard to trace because of inconsistent naming or bad structure. After I organized my naming and commenting, it was much easier to debug and implement new features without sacrificing old ones.

From a design perspective, I wanted to have a simple and readable AR interface. I sketched out some layout ideas and tried to position weather data reasonably in the user's line of sight. While the graphics are adequate, I would improve them further by putting them through readability testing under different lighting conditions and possibly incorporating background panels or contrast-enhancing effects.

One of the issues I'm aware of in the project is occasional delays in data retrieval, especially during low network connectivity. If I had sufficient time, I would have added more comprehensive error management and user notification, such as loading indicators or re-retry requests. I would also like to add forecast animations (e.g., cloud drift, rain effects) and make it possible for users to toggle between current and forecasted weather to improve their experience.

One of the highlights of the project was researching UK standards of weather (i.e., how “heavy rain" or "high cloud cover" are defined) and using that to translate raw numbers into easy-to-understand categories. This made the data more understandable and meaningful to its users.

Overall, this project taught me how to incorporate live data into Unity, gave me more confidence in coding, and taught me more about developing AR experiences that respond to real-world conditions. While there are many ways in which I can still improve, I'm proud of how much I've learned and accomplished from conceptualization through actual implementation.